The theoretical pKa of propanoic acid

Introduction

The acidity of a weak acid can be determined directly by titration of the weak acid with a strong base and measuring the pH with a pH probe. Plotting the pH as the strong base is added volumetrically produces a curve. The pH rises quickly as the strong base is added initially to the solution, but the solution does not have a high enough pH to break the bond to the acidic hydrogen. The curve is then seen to level off and the pH increase of the solution dramatically slows. This happens because the solution, as base continues to be added, is now basic enough to cause the acidic bond to break and thus the base and the weak acid are in equilibrium. This region gives rise to the notion of Ka which is the acidic equilibrium constant for the weak acid and pKa which is just the p (negative log) function of this equilibrium constant. The curve then rises dramatically because the equilibrium region is passed and the solution has become so basic that the weak acid has been completely deprotonated.

These pKa values have been determined for most known weak acids and many tables have been constructed including these experimentally determined values. However, if a new weakly acidic molecule is synthesized in the laboratory, the pKa cannot be easily looked up or determined from established tables. This titration will have to be performed to determine the acidity of said compound. This is somewhat discouraging because titration can be a very drawn out process and is usually done several times in order to establish a standardized result.

An easier way to determine pKa is to base a change in pKa relative to the pKa of a structurally similar molecule. For example, if there is a table that includes the pKa of formic acid which has the formula H(CO)OH, but does not include the structurally similar compound acetic acid CH_3(CO)OH a knowledge of what the methyl group does to the pKa would be useful and could even be quantified. This difference could then be applied to the pKa of formic acid to receive the pKa of acetic acid.

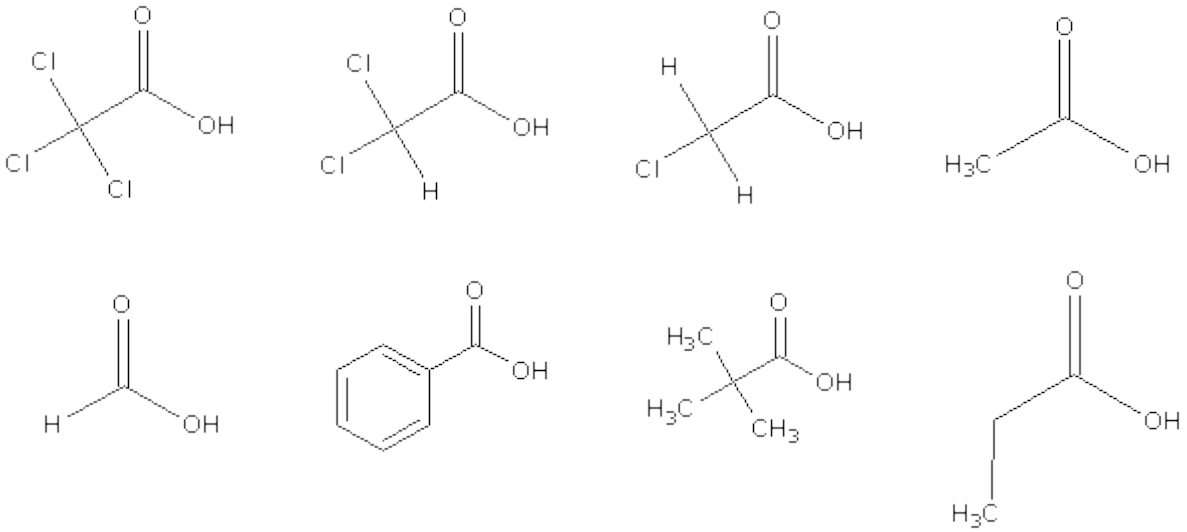
This lab is based on a similar principal as that mentioned in the previous paragraph. A trend is set up among seven structurally similar carboxylic acids by results acquired from calculating the electrostatic potential of the acidic hydrogen atom with the Spartan computational chemistry software package. This trend is then used to calculate the unknown pKa of propanoic acid which is then reported.

The Spartan computational suite of programs uses approximations to the quantum mechanical Schrödinger equation to produce precise and accurate predictions of chemical properties. The Schrödinger equation is a mathematical formula which, if completely solved for a molecule of interest, would be able to glean all information pertaining to the chemical properties of the molecule. The difficulty lies in actually solving the equation which is too computationally complex to be solved and thus approximations are made to solve the equation and extract the information needed. Thus, predictions resulting from these calculations are only as good as the approximations that have been made to make the Schrödinger equation easier to solve.

In this lab, you will be using the Hartree-Fock approximation with a 3-21G* basis set. This approximation is fairly good for most simple property calculations and is fairly quick with the aid of a computer. The basis set used is needed to model atomic orbitals and is used in most quantum chemical applications. The choice of basis set is very important. A larger basis set leads to more accurate predictions, but can make the computations much more time expensive. It is, therefore, best to use the smallest basis set that gives a satisfactory result. The 3-21G* basis set is one of the smallest basis sets available and the consequence of using a smaller basis set should be seen if a larger basis set is chosen to do all said calculations. The basis set and approximation have been chosen for computational efficiency while still gaining a fair amount of accuracy.

You will be calculating the electrostatic potential map of the molecule which is a measure of

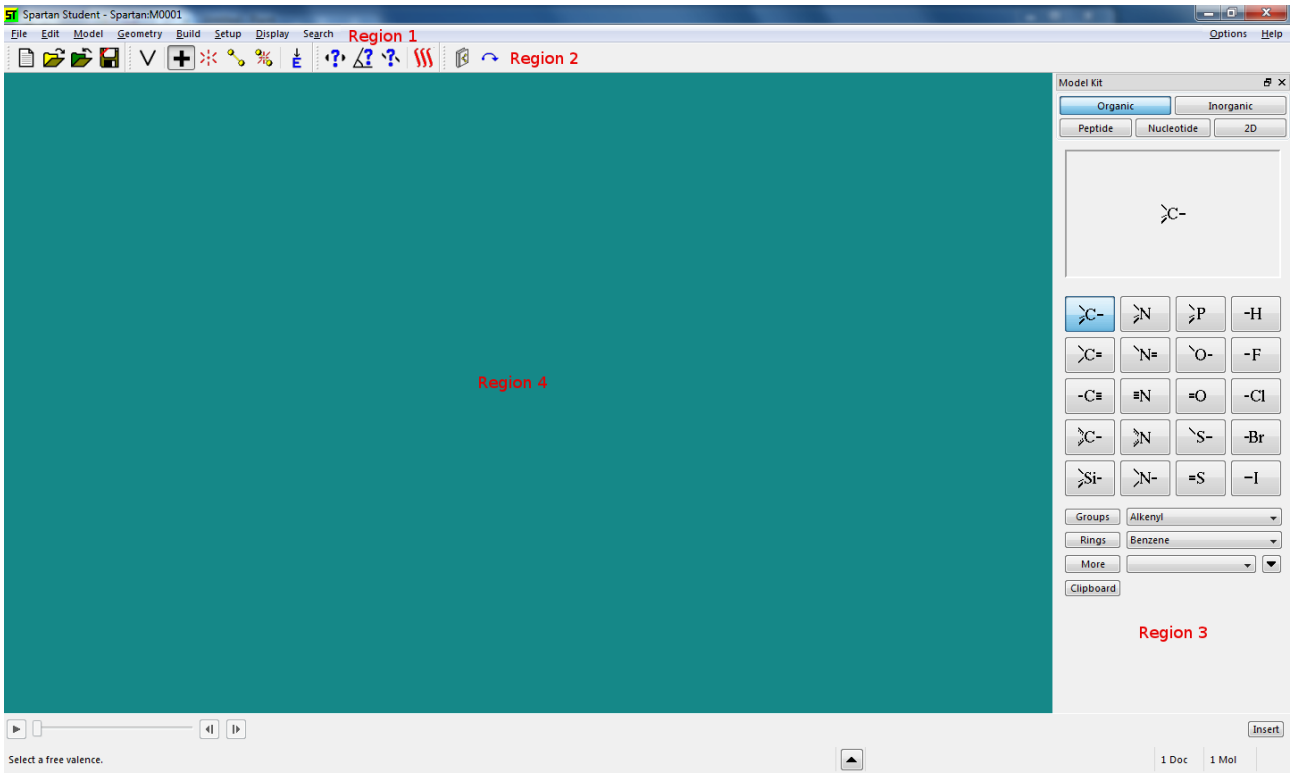
the polarity and charge distribution within a molecule. The electrostatic potential around the acidic hydrogen in a carboxylic acid is very high which corresponds to a large amount of positive charge or an electron deficiency. If more electrons are withdrawn from the bond between the acidic hydrogen and oxygen then the molecule will become more acidic. Thus a more electronegative constituent bonded to the molecule will increase the acidity of that hydrogen. You will be measuring the electrostatic potential of that acidic hydrogen for the seven molecules below: (from first row left to second row right) trichloroacetic acid, dichloroacetic acid, chloroacetic acid, acetic acid, formic acid, benzoic acid, and pivalic acid. These acids are all carboxylic acids and thus all have the structurally similar COOH tail to them. This electrostatic potential will then be plotted against given pKa values to give an equation relating pKa to electrostatic potential. The pKa of propanoic acid (the last acid on the bottom row) can then be determined by just calculating the electrostatic potential of its hydrogen and applying the formula.



Spartan introduction

Before we get started, it is useful to get acquainted with the different areas of the Spartan viewing window. On the next page is a snapshot taken of the Spartan viewing window. I highlight four regions of the Spartan map that I refer to throughout the lab procedure. Region 1 at the top of the snapshot is what I refer to as the upper toolbar and includes the normal dropdown menus of File and Edit etc. Region 2 is right below the upper toolbar with clickable buttons that have descriptive depictions of their function in the program, I refer to this region as the lower toolbar. Region 3 is where most of the molecule building will take place and conveniently it has already been named the model kit; I will refer to it as such. Region 4, where all the changes to the molecule are visualized, I will refer to as the main window. This region is the biggest part (teal) part of the overall viewing window.

Now before we begin the procedure of the experiment let me take you on a quick and dirty tour of the four regions I have highlighted. I will not explain how everything works on my tour, but instead give you essential information that is necessary to complete the lab in an efficient manner. Also it is important to note that the snapshots included in this lab may be using a different version of Spartan than the one you'll be using. Please keep that in mind.



Upper toolbar (Region 1)

In the upper toolbar, we have the familiar File and Edit drop down menus. The File menu creates new files, closes and opens old ones, and saves your progress. The Edit menu can undo, redo, cut, copy, and paste, etc. The Model menu manipulates how your molecule is visualized (ball and stick, tubes, space filling...) and is not necessarily used in this lab. The Geometry and Build menus have options to manipulate the geometry and structure of your molecule and have similar options to what you can use in the lower toolbar. I will go over these options necessary for this experiment when I go over the lower toolbar. There are three options in the Setup menu of which only the options calculation and submit are needed for this experiment. You will use this option to set up and run the specific method of calculation. The Display menu will be used to display different properties and maps produced in experiments. You will mostly be using this for the properties and surfaces options. The Search menu is reserved for more complicated structures and will not be used here.

Lower Toolbar (Region 2)

The Lower toolbar has buttons that are rather self-explanatory. The first four buttons are: New file, Open file, Close file, and Save. The fifth button is the view molecule (depicted with a big "V") which will hide the model kit region of Spartan and allow you more freedom to zoom, rotate, and move your molecule. The sixth button is the add fragment (depicted with a plus) button in which the fragment selected in the model kit can be added into your molecule. The seventh button deletes a selected atom (depicted with a yellow sphere surrounded by six outward pointing red marks). The eighth button makes bonds between two selected fragments (depicted with two yellow spheres connected by a line). The ninth button will delete a bond selected in the molecule (the same depiction as the previous button but with a red x between the two yellow spheres). The tenth button is the quick energy minimization button which attempts to clean up the structure of you molecule (depicted as an E with a down arrow over it). Buttons eleven-thirteen can report and allow the

modification of bond length, bond angle, and dihedrals (the rotational angle between two atoms spaced two atoms apart) respectively. The rest of the buttons are not needed for the beginner using Spartan and will not be used in this lab.

The Model Kit (Region 3)

The Model Kit has five categories inside of it (found underneath the label “Model Kit”): Organic, Inorganic, Peptide, Nucleotide, and Substituent. Only Organic and Inorganic will normally be used. Organic is the default category highlighted with the sp3 or tetrahedral carbon darkened. The darkened fragment in the model kit is the fragment that will be added to the molecule upon insertion and can also be seen in the fragment viewing window (located between the category buttons and the fragment buttons vertically). You will note there are four options for carbon geometries: sp3, sp2, sp1, and a resonance structure carbon. There are also different geometries for the other atoms contained in this category. You should keep in mind the best fragment to build your molecule when using the model kit. In the inorganic category, you will have to choose which element to add and which geometry to use for said element.

Main window (Region 4)

I will conclude the tour with a few pointers needed in Spartan to use it effectively. In order to rotate the molecule you can left click anywhere in the main window and move up and down to rotate vertically and side to side to rotate horizontally. To move your molecule, right click on the molecule in the main window and move it to where you want. To zoom in or zoom out you can use the middle mouse button or hold control and left click. Finally, when you choose to create a new file for a new calculation, be sure to save and close the file you have finished. If you do not do this, your data in the file may be lost or you could end up with several molecules in the same file. Also, any atom addition can always be undone with the delete atom button. It is very important to remember that there are delete functions in the program. This is because, as of writing this, undo has been disabled in the student version (I'm not sure why).

This is all of the information I will include on my Spartan tutorial. A more comprehensive user manual is available at the manufacturer's website. The link is posted below if you wish to use it.

http://www.wavefun.com/products/windows/Student/win_student.html

Lab procedure

Building the molecules

Formic acid

1. Start with creating a new file by either clicking the first button on the lower toolbar or selecting new from the file dropdown menu in the upper toolbar.
2. First let's start by making the carboxylic acid group since we will make it a total of eight times throughout the experiment. Select the (trigonal planar) sp2 carbon from the model kit which should be right below the sp3 carbon (which is highlighted by default). The sp2 carbon should be in the 1st column and 2nd row under the fragment viewing window. Click anywhere in the main viewing window and this carbon will be inserted. You should see a grey sphere with three yellow spokes poking outward. These spokes will be called free valences throughout the rest of this explanation. One of these free valences is shorter and seems to have a slit up the middle of it. This is our double bond valence.
3. Now click the double bonded oxygen which is in the 3rd row and 3rd column in the model kit. If you now click the double bonded valence on the sp2 carbon you will create a carbonyl bond.
4. Next, click the divalent oxygen which is in the 2nd row and 3rd column of the model kit. If you click one of the other two valences the carboxylic acid fragment is complete. The remaining free valences will be assumed to be hydrogen unless they are exchanged for another atom. You have thus also completed the build of formic acid.
5. Instead of doing a calculation here we are going to build all seven of our reference molecules and do a batch calculation on them all. Go to File in the upper toolbar and select new molecule. You will see formic acid disappear, but you did not erase it. It's just in the background.

Acetic acid

1. Start building acetic acid by following steps 2-4 in the last procedure and build carboxylic acid functional group.
2. Now choose the sp3 carbon in the model kit which should be in the first row and first column under the fragment viewing window. Then click your last free valence on your carbonyl carbon to add the methyl fragment.
3. Now, you have built acetic acid. Choose new molecule from the file menu in the upper toolbar.

Pivalic acid

1. To build Pivalic acid build acetic acid as you did in the last procedure.
2. Now choose the sp3 carbon again (first row and first column) and click each of the three free valences on the sp3 carbon that is attached to the carbonyl.
3. You have built pivalic acid. Choose new molecule from the file menu in the upper toolbar.

Chloroacetic, Dichloroacetic, and Trichloroacetic acid

1. Build acetic acid.
2. Now replace all three valences on the sp3 carbon with the chlorine atoms from the model kit (3rd row and 4th column)
3. You have built trichloroacetic acid. Now choose new molecule from the file menu.

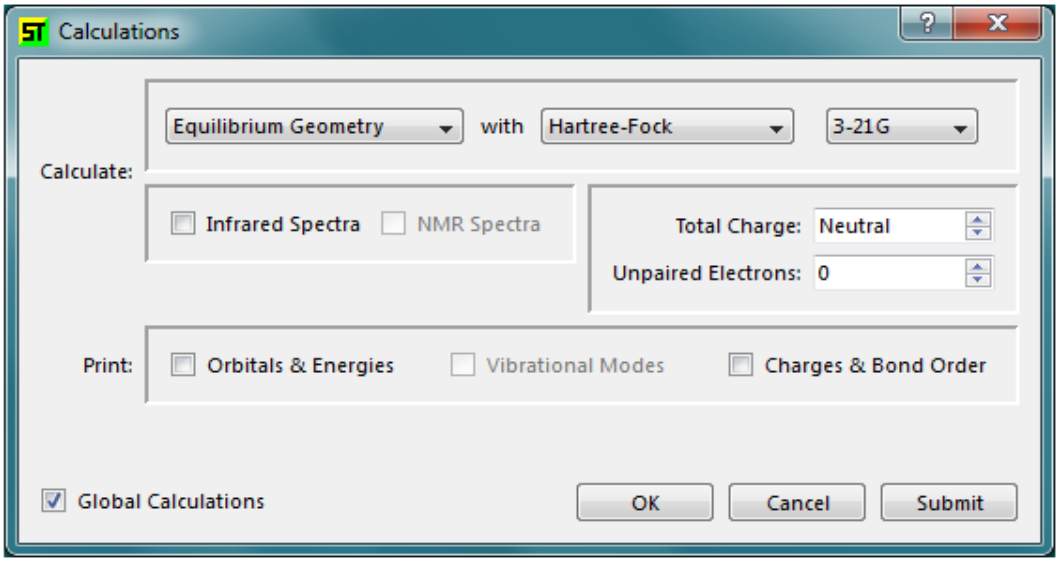
4. Repeat step 1 and 2, but replace only two of the free valences on the sp3 carbon with chlorines.
5. You have built dichloroacetic acid. Choose new molecule in the file menu.
6. Repeat step 1 and 2, but replace only one of the free valences on the sp3 carbon with chlorine.
7. This is chloroacetic acid. Now choose new molecule from the file menu once again.

Benzoic acid

1. Build the carboxylic acid functional group as done six other times. Do not make acetic acid by adding the sp3 carbon.
2. Now in the model kit at the bottom there are buttons listed in this order from top to bottom: Group, Rings, More, and Clipboard. Click rings and then use the drop down menu to click benzene. Once this is done a benzene ring should be shown in the fragment viewing window.
3. Click the remaining free valence on the carbonyl carbon and an entire benzene ring will be added to you molecule.
4. You have now built benzoic acid which is the last of the molecules in this calculation. Do not select new molecule this time.

<u>Performing the calculations</u>

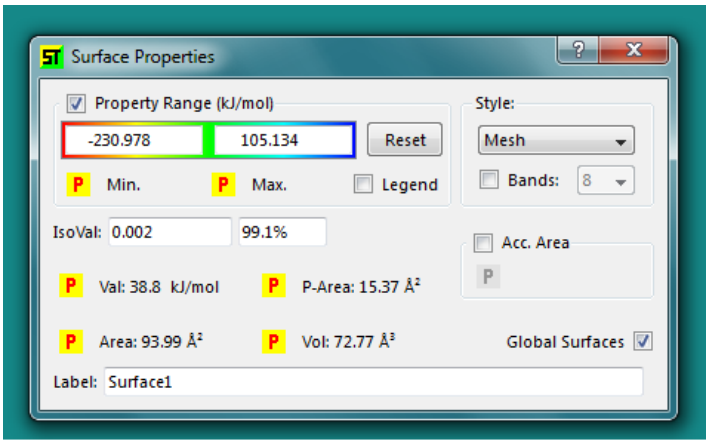
1. Now that all of your molecules are built you can put pre optimized structures for your molecules into the viewing window using Spartan's database library. If you look down out the bottom of the main viewing window, an arrow can be found next to a window that says the name of the current molecule showing. Click this and select from the menu shown in the snapshot the 3-21g option. Then click replace all to replace all of the molecules you built with database structures.
2. Now select setup from the lower toolbar and select calculations. You will receive a pop up window like the one shown in the snapshot. Make sure all settings are set correctly. You



ibrium geometry calculation with Hartree Fock using 3-21G*. Also at the bottom of the box make sure global calculations is checked

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3. You will see a pop up that will ask you to save your molecule. Just save as carboxylic acids and then the calculation will start. You will receive another pop up letting you know when the calculations finish.
4. Now go up to the lower toolbar and select display and choose spreadsheet. You will see a spreadsheet appear in a popup with six labeled rows (M001, M002,...) representing your molecules. Right click each box and choose replace with SMD in the popup menu. This will replace the current label with the name of the molecule. Doing this in turn will show all molecules overlapping in the center of the screen.
5. Now go to the upper toolbar and select geometry and uncheck coupled. Now all the molecules can be separated on the screen. This can be done by clicking on one of the overlapping molecules and the holding the right mouse button and dragging to move the selected molecule. Separate all molecules.
6. Now select setup from the upper toolbar and choose surfaces. You will receive a pop up window like the one shown in the snapshot. Choose Add (or more surfaces) and choose density for the surface, potential for the property, and normal (medium) for the resolution. Then click apply and close the surface box. Now select setup from the upper toolbar and choose submit. When the calculation is complete bring up the surfaces pop up as before and check the surface you have calculated. You should see all surfaces on all molecules. Click a molecule and a yellow cursor will appear where you clicked.



7. Now go to display in the upper toolbar and choose properties. This will give you a surface properties pop up like the one shown. Next to the word max shown there is a yellow box with a red letter P. Click this box and all maximum charge distribution values will be shown on the spreadsheet.
8. Now go back to the spreadsheet via the display menu in the upper toolbar and you should see the second column filled up with the value previously copied. Now click the top of the third column and rename it pKa. Now you can insert the following pKa values for the

molecules: Trichloroacetic-0.7; Dichloroacetic-1.48; Chloroacetic-2.85; Acetic-4.75; Formic-3.75; Benzoic-4.19; Pivalic-5.03.

9. After the information is entered, go to the display menu in the upper toolbar and choose plots. A pop up like the one shown will appear. Choose pKa for the x value and the Property Max (Surface 1) for the y value. If you click OK the plot will be shown on the middle of the screen.
10. Now click a plot point and go to display and properties. This will come up with a curve properties pop up. Choose linear fit LSQ in the Fit dropdown box and a linear fit will be shown along with an equation. Note this equation as you will use it to solve for the pKa of propanoic acid.
11. Save and close this file.

Propanoic acid

1. Create acetic acid via the procedure shown earlier in the lab. Once this is done, be sure the tetrahedral carbon is selected and click a free valence on the tetrahedral carbon in acetic acid.
2. The preferred conformation of propanoic acid the same as acetic acid. This is shown below with the acidic hydrogen closer to the carbonyl oxygen. This is not done by default and has to be changed. Select the 13th button in the lower toolbar which should be the question mark third question mark button from the left. Now select the carbonyl oxygen, carbonyl carbon, acid oxygen, and acidic hydrogen sequentially and a number will appear in the lower right hand corner. Change this number to zero and hit enter.
3. Now submit the equilibrium geometry calculation like before and also set up the surface calculation as done in steps 3-4 and 6 from the following procedure. Now bring up the surface as in step seven and click the surface on propanoic acid. Choose display and properties from the upper toolbar and record the number in the red box at the top of the pop up. This is your max number.
4. Use your equation from the procedure and this number to calculate the pKa for propanoic acid.

Data Sheet

What is the formula calculated that relates pKa to maximum charge distribution?

_____

What is x and what is y in this equation?

_____

What is the calculated pKa of propanoic acid?

_____

Questions

Are there any data points that seem to not follow the trend line? What reasons could there be for this? (hint: How are these molecules different from the others?)

Follow up with a larger basis set

This part of the lab has already been performed and the results shall be discussed herein. These calculations were performed with a larger basis set and a different approximation. The trend line is shown like that of the Spartan output received. It may be noted that in the process of doing this lab the data points were very well correlated with pKa, but benzoic acid seemed to be the one outlier of this data set. This was because the simplistic approximation and the smaller data set could not accurately describe the aromatic ring and its effect on the electrostatic potential. It can be seen from this new data that the line is better fitted with benzoic acid no longer being an outlier. The calculation though took a lot longer to finish and so is not required to be repeated.

This result illustrates what is said before about the approximations to the Schrödinger equation. The results are only as good as the approximation and the size of the basis set used. It also illustrates the fact that although computational chemistry is a good tool for predicting chemical trends and explaining results that otherwise could not be easily explained by experiment, one needs to understand the downfalls of their own approximations. The results of computational chemistry calculations, although extremely useful in some cases, are only meant as a companion and not a replacement for experimental data.